

Spring 24

Problem: Gives $\log_{10} 654 = 2.8156$, $\log_{10} 658 = 2.8182$, $\log_{10} 659 = 2.8189$, $\log_{10} 661 = 2.8202$. Find $\log_{10} 656$ by using Lagrange method.

Solⁿ: For the values x_0, x_1, x_2, x_3 , Lagrange's formula is

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$x = 656$ Here, $x_0 = 654, y_0 = 2.8156, x_1 = 658, y_1 = 2.8182, x_2 = 659, y_2 = 2.8189, x_3 = 661, y_3 = 2.8202$

$$\begin{aligned} y(656) &= \frac{(656-658)(656-659)(656-661)}{(654-658)(654-659)(654-661)} \times (2.8156) + \\ &\frac{(656-654)(656-659)(656-661)}{(658-654)(658-659)(658-661)} \times (2.8182) + \\ &\frac{(656-654)(656-658)(656-661)}{(659-654)(659-658)(659-661)} \times (2.8189) + \\ &\frac{(656-654)(656-658)(656-659)}{(661-654)(661-658)(661-659)} \end{aligned}$$

$$P(656) = \frac{(-2)(-3)(-5)}{(-1) \times (-4) \times (-7)} \times 2.8156 + \frac{(2)(-3)(-6)}{4 \times (-1) \times (-3)} \times 2.8182$$

$$+ \frac{(2)(-2)(-5)}{(5)(1)(-2)} \times 2.8189 + \frac{(2)(-2)(-3)}{(7)(3)(2)} \times 2.8202$$

$$= 0.6033 + 7.0455 - 5.6378 + 0.8058$$

$$= 2.8168$$

Hence, the estimated value of $\log_{10} 656 = 2.8168$

2010 Problem: certain corresponding values of x and $\log_{10} x$ are $(300, 2.4771)$, $(304, 2.4829)$, $(305, 2.4843)$, and $(307, 2.4871)$. Find $\log_{10} 301$ by using Lagrange's formula.

Solⁿ: For the values x_0, x_1, x_2, x_3 , Lagrange's formula is

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} f_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} f_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} f_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} f_3$$

Here, $x = 301$,

$$x_0 = 300 \quad f_0 = 2.4771$$

$$x_1 = 304 \quad f_1 = 2.4829$$

$$x_2 = 305 \quad f_2 = 2.4843$$

$$x_3 = 307 \quad f_3 = 2.4871$$

$$y = \frac{(301-300)(301-305)(301-307)}{(300-304)(300-305)(300-307)} \times (2.4771)$$

$$+ \frac{(301-300)(301-305)(301-307)}{(304-300)(304-305)(304-307)} \times (2.4829)$$

$$+ \frac{(301-300)(301-304)(301-307)}{(305-300)(305-304)(305-307)} \times (2.4843)$$

$$+ \frac{(301-300)(301-304)(301-305)}{(307-300)(307-304)(307-305)} \times (2.4871)$$

$$= \frac{(-3)(-4)(-6)}{(-9)(-5)(-7)} \times (2.4771) + \frac{1 \times (-4)(-6)}{4 \cdot (-1) \cdot (-3)} \times (2.4829)$$

$$+ \frac{1 \times (-3)(-6)}{5 \cdot 1 \cdot (-2)} \times (2.4843) + \frac{1 \cdot (-3)(-4)}{7 \cdot 3 \cdot 2} \times (2.4871)$$

$$= 1.2739 + 4.9658 - 4.4717 + 0.7106$$

$$= 2.4786$$

Hence the estimated value of $\log_{10} 301 = 2.4786$

(Ans.)

$$p \cdot \frac{(p-x)(q-x)(r-x)}{(p-1)(q-1) \cdot 1} + q \cdot \frac{(p-x)(q-x)(1-x)}{(p-0)(q-0)(1-0)} = \beta$$

$$p \cdot \frac{(q-x)(1-x) \cdot x}{(q-p) \cdot (1-p) \cdot p} + q \cdot \frac{(p-x)(1-x) \cdot x}{(p-0) \cdot 1 \cdot 0} = \beta$$

Problem: Find a polynomial by Lagrange's formula and find $y(2)$. Following the data

x	y
$x_0 \rightarrow 0$	3
$x_1 \rightarrow 1$	4
$x_2 \rightarrow 3$	30
$x_3 \rightarrow 4$	67

Soln: For the values x_0, x_1, x_2 and x_3 Lagrange's formula

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

Here, $x_0=0, y_0=3$
 $x_1=1, y_1=4$
 $x_2=3, y_2=30$
 $x_3=4, y_3=67$

$$\therefore y = \frac{(x-1)(x-3)(x-4)}{(0-1)(0-3)(0-4)} \cdot 3 + \frac{(x-0)(x-3)(x-4)}{1 \cdot (1-3)(1-4)} \cdot 4 + \frac{x(x-1)(x-4)}{3 \cdot 2 \cdot (3-4)} \cdot 30 + \frac{x(x-1)(x-3)}{4 \cdot (4-1) \cdot (4-3)} \cdot 67$$

$$y = \frac{(x^2 - 3x - 2 + 3)(x-4)}{-4} + \frac{x(x^2 - 4x - 3x + 12)}{(-2)(-3)} + \frac{2}{(-1)} + \frac{x(x^2 - 4x - x + 4)}{-5}$$

$$+ \frac{x(x^2 - 3x - x + 3)}{4(3) \cdot 4} \cdot 67$$

$$= \frac{(x^2 - 4x + 3)(x-4)}{-4} + \frac{2x(x^2 - 7x + 12)}{3} - \frac{5x(x^2 - 5x + 4)}{5}$$

$$+ \frac{67}{12} x(x^2 - 4x + 3)$$

$$= \frac{x^3 - 4x^2 - 4x^2 + 16x + 3x - 12}{-4} + \frac{2x^3 - 14x^2 + 24x}{3} - (5x^3 - 25x^2 + 20x)$$

$$= \frac{(x^3 - 8x^2 + 19x - 12)}{-4} + \frac{2x^3 - 14x^2 + 24x}{3} - \frac{5x^3 - 25x^2 + 20x}{1}$$

$$= \frac{-3(x^3 - 8x^2 + 19x - 12) + 4(2x^3 - 14x^2 + 24x) - 12(5x^3 - 25x^2 + 20x)}{12}$$

$$= \frac{-3x^3 + 24x^2 - 57x + 36 + 8x^3 - 56x^2 + 96x - 60x^3 + 300x^2 - 240x}{12}$$

$$= \frac{76x^3 - 88x^2 + 75x^3 - 63x^3 + 324x^2 - 324x^2 + 297x - 237x + 36}{12}$$

$$= \frac{12x^3 + 36}{12} = \frac{12(x^3 + 3)}{12} = x^3 + 3$$

$$= \frac{12x^3 + 36}{12} = \frac{12(x^3 + 3)}{12} = x^3 + 3$$

$$\therefore y(2) = 2^3 + 3$$

$$= 11$$

(Ans.)

Sp. No. 24

Problem: Write an algorithm for Lagrange's formula. Find the polynomial which fits ^(given) the following data

$(-1, 7)$, $(1, 5)$ and $(2, 15)$

Soln: For the values x_0, x_1, x_2 , Lagrange's formula

$$y = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} y_2$$

Here,

$$x_0 = -1, \quad y_0 = 7$$

$$x_1 = 1, \quad y_1 = 5$$

$$x_2 = 2, \quad y_2 = 15$$

$$\therefore y = \frac{(x-1)(x-2)}{(-1-1)(-1-2)} 7 + \frac{(x+1)(x-2)}{(1+1)(1-2)} 5 + \frac{(x+1)(x-1)}{(2+1)(2-1)} 15$$

$$= \frac{(x^2 - 2x - x + 2)}{(-2)(-3)} 7 + \frac{(x^2 - 2x + x - 2)}{(2)(-1)} 5 + \frac{x^2 - 1}{3 \cdot 1} 15$$

$$\sqrt[2]{\frac{6,2}{3,1}}$$

$$= \frac{(x^2 - 3x + 2)}{6} - \frac{5(x^2 - x - 2)}{2} + 5(x - 1)$$

$$= \frac{7(x^2 - 3x + 2) - 15(x^2 - x - 2) + 30(x - 1)}{6}$$

$$= \frac{7x^2 - 21x + 14 - 15x^2 + 15x + 30 + 30x - 30}{6}$$

$$= \frac{37x^2 - 15x - 6x + 14}{6}$$

$$= \frac{37x^2 - 21x + 14}{6}$$

$$\frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)}$$

Ans:

$$\frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)}$$

$$\frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)}$$

$$\frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)} + \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x-3)(x-4)}$$

$$\begin{aligned} &= \frac{37x^2 - 21x + 14}{6} \\ &= \frac{37x^2 - 21x + 14}{6} \end{aligned}$$

$$0 = 0$$

$$1 = 1$$

$$18 = 18$$

$$232 = 232$$

$$232 = 232$$

$$0 = 0$$

$$1 = 1$$

$$18 = 18$$

$$232 = 232$$

$$232 = 232$$

Problem: Find $y(2)$ from the following data using Lagrange's formula.

x	0	1	3	4	5
y	0	1	81	256	625

Solⁿ: For the values x_0, x_1, x_2, x_3 and x_4 . We

know Lagrange's formula is

$$\begin{aligned}
 y = & \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} y_1 \\
 & + \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} y_2 \\
 & + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} y_3 \\
 & + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} y_4
 \end{aligned}$$

Here, $x=2$

$$x_0 = 0$$

$$x_1 = 1$$

$$x_2 = 3$$

$$x_3 = 4$$

$$x_4 = 5$$

$$y_0 = 0$$

$$y_1 = 1$$

$$y_2 = 81$$

$$y_3 = 256$$

$$y_4 = 625$$

$$\begin{aligned}
 y &= \frac{(2-1)(2-3)(2-4)(2-5)}{(0-1)(0-3)(0-4)(0-5)} \cdot 0 + \frac{(2-0)(2-3)(2-4)(2-5)}{(1-0)(1-3)(1-4)(1-5)} \cdot 1 \\
 &+ \frac{(2-0)(2-1)(2-4)(2-5)}{(3-0)(3-1)(3-4)(3-5)} \cdot 81 + \frac{(2-0)(2-1)(2-3)(2-5)}{(4-0)(4-1)(4-3)(4-5)} \cdot 256 \\
 &+ \frac{(2-0)(2-1)(2-3)(2-4)}{(5-0)(5-1)(5-3)(5-4)} \cdot 625
 \end{aligned}$$

$$= 0 + \frac{2(4)(-2)(-5)}{1(-2)(-3)(-4)} + \frac{2 \cdot 1 \cdot (-2)(-5)}{3 \cdot 2 \cdot (-1)(-2)} \cdot 81 + \frac{2 \cdot 1 \cdot (-1)(-2)}{4 \cdot 3 \cdot 1 \cdot (-1)} \cdot 256$$

$$= \frac{(x-x)(x-x)(x-x)}{(x-x)(x-x)(x-x)} + \frac{2 \cdot 1 \cdot (-1)(-2)}{5 \cdot 4 \cdot 2 \cdot 1} \cdot 625 \frac{(x-x)(x-x)(x-x)}{(x-x)(x-x)(x-x)} = B$$

$$\frac{(x-x)(x-x)(x-x)}{(x-x)(x-x)(x-x)} + \frac{0.5 + 81 + 128 + 62.5}{(x-x)(x-x)(x-x)} + B$$

(Ans:)

$$\begin{aligned}
 1 &= 0 \\
 2 &= 1 \\
 3 &= 0 \\
 4 &= 0
 \end{aligned}$$

$$\begin{aligned}
 1 &= 0 \\
 2 &= 1 \\
 3 &= 0 \\
 4 &= 0
 \end{aligned}$$

Ans